

Washim Akram

Software Engineer

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Professional Summary

Software engineer with several years of experience maintaining backend applications, investigating production issues, testing software changes, and working with data-driven services in international teams. Technical background includes C, C++, PHP, Java with object-oriented programming, Python, Linux, Git, and Docker. Academic work at the University of Vienna includes C program analysis, algorithms, reverse engineering, control-flow analysis, and LLM-assisted software analysis. Currently completing an M.Sc. in Computer Science with a thesis focused on trust-boundary enforcement for agentic reverse-engineering systems.

Technical Skills

Programming Languages: C, C++, PHP, Java (OOP), Python, SQL

Software Engineering: Object-oriented programming, backend application development and maintenance, algorithms, debugging, technical problem solving

Program Analysis & Research: Static program analysis, control-flow analysis, reverse engineering, code-obfuscation analysis, algorithm identification, LLM-assisted code analysis

DevOps & Tools: Git, GitLab, GitHub, Docker (basic), Postman; knowledge of CI/CD, automated builds and deployments

Architecture & Methods: Knowledge of distributed systems, modern software architecture, Scrum, and agile development concepts

Operating Systems & Infrastructure: Linux, Windows, production systems, system configuration, TCP/IP, IP networking, LAN/WAN, VLAN concepts

Data & Integration: Microsoft SQL Server, MySQL, MongoDB, JDBC, Pervasive ETL, SQL stored procedures, XML, XSD/XML Schema, XSLT, XPath, XQuery, JSON, JSON Schema

Professional Experience

Software Engineer

Limerick Resources Ltd.

Jan 2023 – Dec 2023

Dhaka, Bangladesh – Hybrid

- Investigated application and service issues, analysed technical findings, and worked with engineers until problems were resolved.
- Performed structured testing after software changes, validating application behaviour, data handling, and system stability before delivery.
- Worked with XML, XSD/XML Schema, XSLT, XPath, XQuery, JSON, and JSON Schema for data validation, transformation, querying, and backend application integration.
- Configured and troubleshoot JDBC-based database connectivity, including connection and data-access issues in backend applications.
- Coordinated technical work across a team of more than five engineers involved in backend development, testing, and delivery, contributing to an approximately 25% improvement in delivery time through better coordination and structured execution.

Software Engineer

Nassprotech

Mar 2022 – Dec 2022

United States – Remote

- Maintained backend applications and SQL-based services used in daily business operations, with attention to availability, data accuracy, performance, and reliability.
- Investigated production incidents with international engineering colleagues and helped identify application, database, and service-related causes.
- Maintained and optimized SQL stored procedures, data-processing services, ETL workflows, and reporting solutions supporting operational requirements.
- Tested application and service changes after updates and validated system behaviour to support stable production operation.
- Improved technical and operational documentation and contributed to automation-related work for recurring service tasks.

Junior Software Engineer

Nassprotech

Aug 2020 – Mar 2022

United States – Remote

- Supported the development and maintenance of backend applications and data services operating in production environments.
- Diagnosed application, database, and operational issues with senior team members and contributed to resolving technical problems.
- Worked with SQL, ETL processes, and reporting systems while building practical experience with production data workflows and software services.
- Tested application and service changes, supported maintenance activities, and documented completed technical work.

Lab Teaching Assistant

Daffodil International University

Jan 2019 – Apr 2020

Dhaka, Bangladesh – On-site

- Supported students with programming, software-usage, and computer problems during practical laboratory sessions.
- Helped students investigate and resolve object-oriented programming issues by explaining technical causes and practical correction steps.
- Reviewed technical reports and provided feedback on implementation, problem solving, and documentation.

Selected Academic Projects & Research

Algorithm Identification in C Programs Obfuscated Using Control-Flow Flattening

University of Vienna

- Developed a Python-based analysis pipeline for identifying algorithms in C programs after control-flow flattening with Tigress.
- Combined static semantic feature extraction, rule-based classification, and LLM-assisted analysis to examine program behaviour despite transformed control flow.
- Worked with sorting, searching, numeric, and string-processing algorithms and automatically verified results against ground-truth labels stored in JSON.

LLM-Based Reconstruction of Long Function Call Chains in C Programs

University of Vienna

- Built a Python framework to generate C programs and evaluate LLM-based reconstruction of long function call chains.
- Tested direct calls, function pointers, pointer arrays, structure-based function pointers, and switch-based dispatch at depths of up to 1000 functions.
- Implemented automated ground-truth generation, API communication, exact-match verification, failure detection, and JSON result storage.

Education

Master of Science in Computer Science

2024 – Present

University of Vienna, Vienna, Austria

Currently in the final semester.

Master's Thesis: *Trust-Boundary Enforcement for Agentic Reverse Engineering of Obfuscated Binaries: Securing LLM Agents Against Adversarial Artifacts and Tool-Chain Manipulation*

Bachelor of Science in Software Engineering

2016 – 2020

Daffodil International University, Bangladesh

CGPA: 3.95 / 4.00 | German equivalent: 1.1

Languages

English: C1 | German: B1, currently working toward B2 | Bengali: Native

Additional Information

Eligible to work in Austria.